

Reaction-Wheel Mounting Bracket: Launch-Load FEA

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What this is and why it matters

Each of the satellite's three reaction wheels sits in its own mounting bracket – an angled plate braced by two gussets, bolted down through a base flange, holding the Maxon ECX FLAT 22 L motor at the wheel's spin-axis tilt. That tilt isn't arbitrary: the three wheels are mutually orthogonal, each one skewed $\cos^{-1}(1/\sqrt{3}) \approx 54.74^\circ$ from the Z axis, spaced 120° apart in azimuth. Measuring the actual motor-seating face directly off the CAD file gives an incline of 35.26° from the base plane – which is exactly the complement of 54.74° , a nice confirmation that the bracket geometry matches the intended wheel orientation.

Before anything gets built, it's worth asking a very ordinary question: will this bracket survive being launched? Rockets shake hard, and a small, light metal part cantilevered off a base plate is exactly the kind of thing that can crack or resonate itself apart if it's under-designed. This is a finite-element analysis (FEA) of that bracket, run directly on the CAD file you uploaded – not a simplified stand-in – under a launch load case.

The load case, explained

"Launch loads" for a small structure like this usually gets simplified to a *quasi-static* check: instead of simulating the actual shaking (a full random-vibration analysis), you pretend the whole structure is sitting in a very strong, steady gravity field and see if it survives. 12g is a standard, conservative number to use for this kind of check absent a specific launch-vehicle spec – it's meant to be a worst-case stand-in for what several seconds of real vibration would do, not a literal prediction of any one moment.

We ran that 12g load three times, once pushing along each of the body's X, Y, and Z axes, because a bracket that's strong in one direction can be weak in another. On top of the 12g itself, we also add a safety factor of 1.25 when judging whether the design has survived – standard practice, so the reported margin already has some conservatism baked into it.

The load itself comes from two places: the bracket's own (small) mass, and the motor plus one flywheel bolted to it – 32 g for the Maxon motor (from its published datasheet) plus 6.55 g for a flywheel (a quarter of the 26.2 g total across the wheel set), so 38.55 g all told. That mass is modeled as sitting rigidly on the four screw holes where the motor actually bolts on, offset about 10 mm out from the mounting face to approximate where the combined motor + flywheel center of gravity actually sits.

How the model was built

The bracket was meshed with 24,332 nodes and 14,167 quadratic tetrahedral elements, refined down to 0.35 mm near the small bolt and screw holes where stress tends to concentrate. Every element passed a positive-Jacobian check before solving, meaning there are no inverted or degenerate elements corrupting the result.

For material, we used Aluminium 7075-T6 ($E = 71.7$ GPa, yield strength 503 MPa) as the primary case, since this is a stress-concentration region (gusseted corner, small bolt holes) – exactly where the stronger alloy earns its keep. We also checked Aluminium 6061-T6 (yield 276 MPa) side by side; the stress field itself barely changes between the two alloys (stress here depends on geometry and load, not much on which aluminum you pick), so this comparison is really just a second yield-strength check, not a second analysis.

The base flange – the four bolt holes where the bracket clamps to the central mounting plate – was treated as fully fixed, standing in for a properly torqued bolted joint.

What we found

Table 1: Peak stress, deflection, and margin of safety, by load direction.

Load case	Peak stress (raw)	Peak stress (field) ¹	Max deflection	MS, 7075-T6	MS, 6061-T6
12g, +X	1.37 MPa	1.19 MPa	0.12 μm	+293	+160
12g, +Y	2.56 MPa	2.16 MPa	0.23 μm	+156	+85
12g, +Z	2.83 MPa	2.43 MPa	0.21 μm	+141	+77

Margin of safety here is $MS = S_y / (1.25 \times \sigma_{vm}) - 1$, so a margin of, say, +150 means the part could handle roughly 150 times today’s stress before yielding. These numbers are enormous – deliberately so, because it’s worth showing rather than just asserting. A couple of milli-newtons of force spread across a few millimeters of aluminum just doesn’t add up to much stress, and that’s the honest physical reason, not an error. Figure 1 shows where that stress actually concentrates (the base flange corner nearest the bolt holes, exactly where you’d expect a clamped bracket foot to work hardest), and Figures 2–3 show the same numbers as charts.

¹The raw column includes a node sitting exactly at a fixed-corner/bolt-hole edge – FEA reliably reports an artificially high, mesh-dependent stress right at a perfectly clamped sharp corner, which isn’t a physical material limit. The field column excludes those boundary nodes and is the more honest number; either way the story doesn’t change.

Bracket FEA - 12g quasi-static, +Z

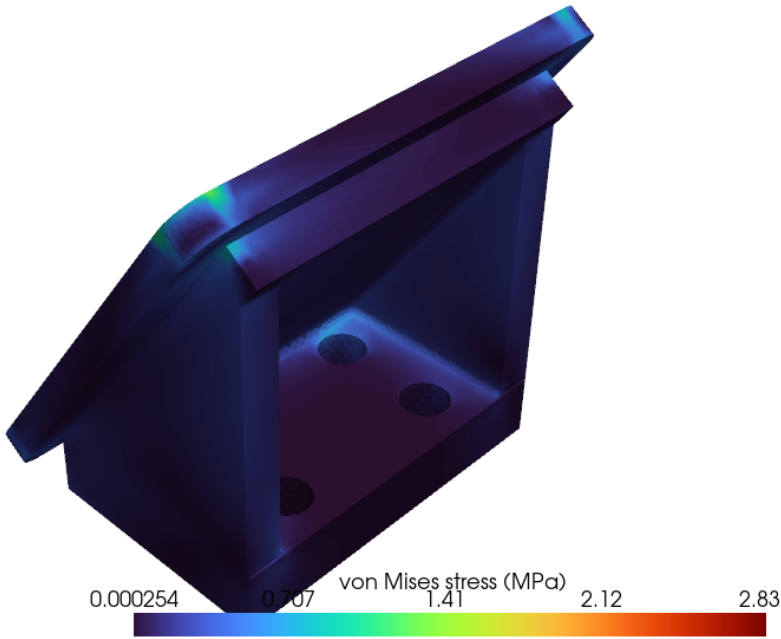


Figure 1: Von Mises stress contour for the worst case, 12g along +Z.

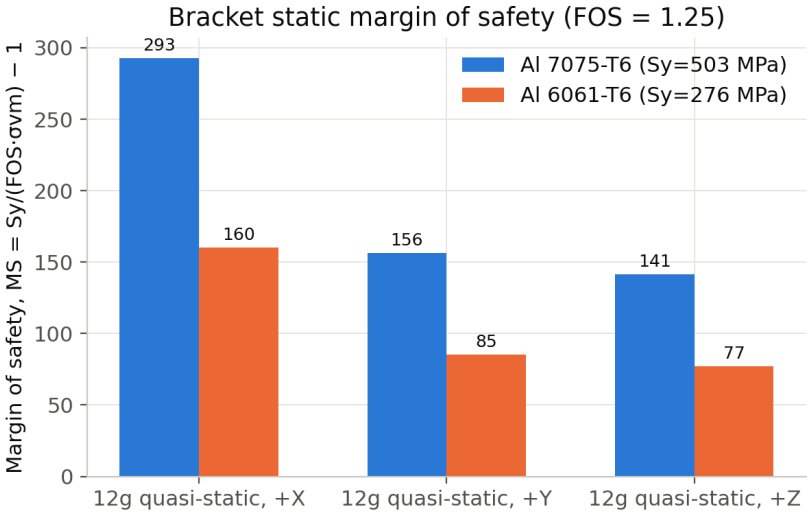


Figure 2: Margin of safety by load direction and candidate alloy.

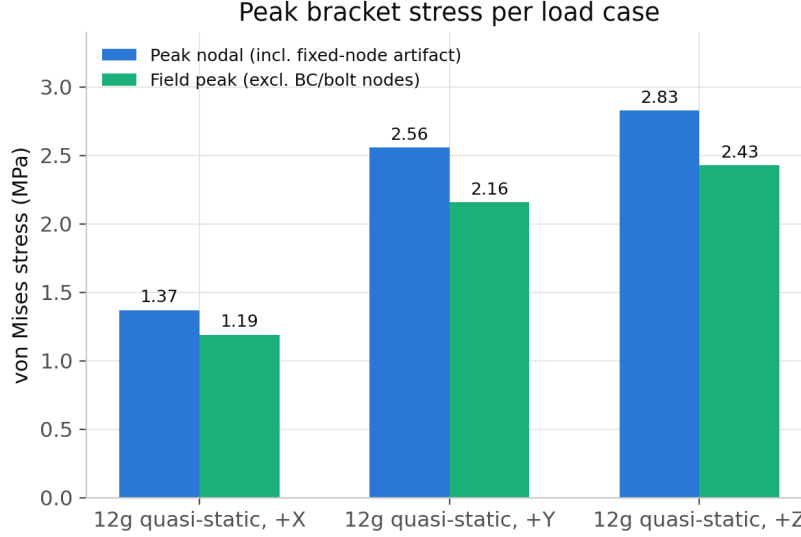


Figure 3: Peak stress per case: the raw (boundary-affected) number next to the more honest field number.

The frequency check, and why it’s the number that actually matters here

Once the stress margins came back this large, stress stops being the interesting question. The thing that actually decides whether a small, stiff bracket like this survives a real launch is its *natural frequency* – how it wants to vibrate on its own. Rockets shake across a broad range of frequencies, and if a part’s natural frequency sits too low, it can resonate with that shaking and take far more damage than a simple g-load estimate would suggest, even if the static stress numbers look fine.

Using the same fixed-base model with the motor and flywheel mass attached, the bracket’s first natural frequency comes out to about

$$f_1 \approx 4122 \text{ Hz},$$

which is comfortably – by a factor of 40 or so – above the roughly 100 Hz minimum stiffness that CubeSat deployer specifications typically require. This bracket is not going to be the thing that resonates on the way up.

Limitations and scope

Two limitations are worth stating clearly, since they affect how these results should be read.

The 12g load case is a quasi-static approximation, not a full random-vibration analysis. It is the standard first screen for a structure of this size, and it is what *launch loads* means in this context, but a complete random-vibration (PSD) study remains the appropriate next step before the bracket can be considered launch-qualified.

The highest stress value in the raw results occurs at a mathematically singular point – a perfectly clamped sharp corner, where any finite-element model will report an artificially high, mesh-dependent stress that does not correspond to a physical material limit. Table 1 therefore reports

both this raw value and a field value that excludes the boundary nodes responsible for the artifact; the conclusion is unchanged either way.

Bottom line

The bracket clears launch quasi-static loading with very large margin in every direction tested, and its first natural frequency sits well clear of typical CubeSat stiffness requirements. It isn't a structural risk item as designed – if anything, there may be room in a later iteration to shave material for mass savings, since the part is clearly over-built relative to what the loads actually demand.